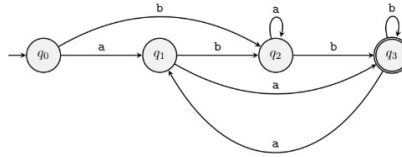


Solve problems 1 through 6. Problem 7 is optional.

Problem 1: Converting DFA to Regular Expressions (10 points)

Convert the following DFA into an equivalent regular expression using the state elimination method. First eliminate q_1 , then q_2 and so on. You must show work.



Problem 2: Designing Context-Free Grammars (5 points)

Solve the following problems.

- (a) Give a context-free grammar for the following language. (2 points)

$$A = \{w \in \{0, 1\}^* : w \text{ contains at least two 0s}\}$$

- (b) Give a context-free grammar for the following language. (3 points)

$$L = \{w \in \{0, 1\}^* : w = 0^3v1^{2i} \text{ where } v \in A \text{ and } i \geq 0\}$$

Problem 3 : Derivations, Parse Trees, and Ambiguity (5 points)

$$S \rightarrow 0S1 \mid 1S0 \mid SS \mid 01 \mid 10$$

Take a look at the grammar above and solve the following problems.

- (a) Show that the grammar above is ambiguous by demonstrating two different parse trees for 011010. (4 points)
(b) Find a string w of length six such that w has exactly one parse tree in the grammar above. (1 point)

Problem 4: Chomsky Normal Form (10 points)

Convert the following grammar into Chomsky Normal Form. You must show work.

$$\begin{aligned} S &\rightarrow a \mid aA \mid B \\ A &\rightarrow aBB \mid \varepsilon \mid D \\ B &\rightarrow Aa \mid b \\ D &\rightarrow \varepsilon \end{aligned}$$

Here a , a , and b are terminals and the rest are variables.

Problem 5: The CYK Algorithm (10 points)

Use the CYK algorithm to determine whether the string $abccccc$ can be derived in the following grammar. You must show the entire CYK table.

$$\begin{aligned} S &\rightarrow AB \\ A &\rightarrow CD \mid CF \\ B &\rightarrow c \mid EB \\ C &\rightarrow a \\ D &\rightarrow b \\ E &\rightarrow c \\ F &\rightarrow AD \end{aligned}$$

Here, a , b , and c are terminals and the rest are variables.

Problem 6: Constructing Pushdown Automata (5 points)

Construct a pushdown automaton for the following language.

$$L = \{w\bar{w}^R : w \in \{0, 1\}^*\}$$

Here, $\bar{w}^R$ denotes the reverse complement of the string w .

For example, $01001101 \in L$ because the second half of the string, 1101, is the reverse complement of the first half, 0100, i.e. $1101 = 0100^R$.

Problem 7: (Bonus) Deja Vu (5 points)

(Note that this is a bonus problem. Attempt it only after you are done with everything else. Even if you do not attempt it, you can get a perfect score. So, do not worry if you find it too hard!)

Take a look at the following languages. They should seem familiar.

$$L_1 = \{w \in \{0, 1\}^* : w = 0^m1^n \text{ where } m \text{ and } n \text{ are both even}\},$$

$$L_2 = \{w \in \{0, 1\}^* : w = 0^m1^n \text{ where } m \text{ and } n \text{ are both odd}\}.$$

We will use these two languages to define the following language.

$$L = \{w \in \{0, 1\}^* : w = uv, \text{ where } u \in L_1, v \in L_2, \text{ and } |u| = |v|\}$$

- (a) Write down a string $w \in L$ such that w is of length eight. (0 point)
(b) Construct a pushdown automaton that recognizes L . Describe what your automaton is doing in two or three sentences. (5 points)